

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (CL)

May 2012

CL309 Construction Planning and Management

Date: 10.05.2012, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a)** What are the duties of an engineer with respect to construction management? **04**
- (b)** State basic difference between CPM and PERT techniques. **03**
- Q - 2 (a)** What is the importance of resource smoothening? Explain the method of resource smoothening with a suitable example. **06**
- (b)** For a project, following activities are evolved. Workout **08**
1. Critical path.
 2. Estimated time to complete the project.
 3. Probability of completion of project in 40 days if Z factor at 93% is 1.5 and at 96% is 1.75.

Activity	Optimistic time in weeks T1	Most likely time in weeks T2	Pessimistic Time in weeks T3
1-2	2	5	8
2-5	4	9	20
2-4	4	7	16
2-3	3	11	20
3-4	0	0	0
4-6	7	10	13
3-7	3	5	13
5-6	3	7	17
6-7	2	3	10
7-8	2	4	6

OR

- Q - 2 (a)** Explain direct cost and indirect cost of project. State the different expenditure comes under it. Discuss their relationship with time using graphs. **06**
- (b)** For the work involving 9 activities with the given data in table below, calculate the following: **08**
1. Standard deviation and work variance.
 2. Estimated schedule time.
 3. Calculate TF, FF and IF for activities 3-5, 6-7, 7-8.

Activity	Optimistic time in weeks T1	Most likely time in weeks T2	Pessimistic Time in weeks T3
1-2	3	6	15
1-6	2	5	14
2-3	6	12	30
2-4	2	5	8
3-5	5	11	17
4-5	3	6	15
6-7	3	9	27
5-8	1	4	7
7-8	4	19	28

- Q - 3 (a) What is Gantt chart? Give reasons why the Gantt chart fails in case of large construction project? 04
- (b) Referring below table, workout optimum duration of project and corresponding minimum cost. Draw CPM network. 05

Activity	Normal duration in day	Normal cost Rs.	Crash period in day	Crash cost Rs.
A 1-2	7	7000/-	3	14500/-
B 1-3	9	4000/-	5	8500/-
C 2-3	4	6000/-	1	9000/-
D 2-4	5	8000/-	3	15000/-
E 3-4	5	5000/-	3	11000/-

- (c) What are the factors to be considered while selection of an equipment? 05
- Q - 3 (a) Enumerate general rules to be followed in case of network diagram. 03
- (b) Refer the following table & prepare the network diagram. Decide the completion period and complete the CPM schedule. 06

Activity Items	Duration in Days	Activity immediately	
		Preceding	Following
A	3	None	B,C
B	2	A	D,E
C	3	A	E
D	5	B	F
E	6	B,C	F,G
F	5	D,E	None
G	4	E	None

- (c) A construction equipment was purchased at Rs. 4,50,000/-. Assuming its salvage value at the end of 5 years to be 12% of purchase cost. Determine book value of each year by Sinking fund method at a rate of 4%. 05

SECTION - II

- Q - 4 Explain various scope and objective of construction management. 07

- Q - 5 (a)** What is the role of executive engineer as part of project management? **04**
- (b)** Explain the need of organization for any project. Mention types of organization and draw diagram of organization which is suitable for construction industries. **05**
- (c)** What are the causes of failures in construction project? How construction management helps to overcome it? **05**

OR

- Q - 5 (a)** Explain the principle of storing and staking of various material and resources on site of construction project. **04**
- (b)** Describe principle of organization in detail. **05**
- (c)** Explain factors affecting various stages of construction management. **05**
- Q - 6 (a)** Write a short note on scraper including its classification, name of its parts, operation in one cycle, advantages and disadvantages. **07**
- (b)** Define rolling resistance, drawbar pull and coefficient of traction. Explain importance of them while selection of construction equipment. **05**
- (c)** What is personnel management? State its function. **02**

OR

- Q - 6 (a)** Write a short note on soil compacting equipment including mode of compaction, types and selection criteria for compaction equipment with their suitability for various soil conditions. **07**
- (b)** How the effect of slope, altitude and temperature condition of site is incorporated in performance calculation of equipment? **05**
- (c)** State the factors affecting efficiency and performance of Indian labour. **02**
